Sentiment Analysis of Digital Customer Service Helpdesk Tickets: A Comparison between BERT and SVM

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Jayusman ; Riyanto Jayadi All Authors

160
Full
Text Views

Abstract

Document Sections

I. Introduction

II. Materials and Methods

III. Methodology

IV. Results and Discussion

V. Conclusions

Authors

Figures

References

Keywords

Metrics

More Like This

Abstract:
This research aimed to perform a comparative analysis of the effectiveness of customer support systems and the management of helpdesk tickets, which play a pivotal role in maintaining customer satisfaction and loyalty in the current digital age. As helpdesk tickets surge in number, leveraging automation to accurately gauge and address customer emotions becomes essential. This research introduces the application of Bidirectional Encoder Representations from Transformers (BERT), a sophisticated natural language processing (NLP) technique, for analyzing the sentiments expressed in helpdesk tickets of digital products. A BERT model was crafted and trained to categorize the sentiments of these tickets as either positive, negative, or neutral. This endeavor aims to equip businesses with deeper insights, enabling them to improve their approach to customer service. The model underwent training with a collection of helpdesk tickets from a variety of digital products, such as software, mobile applications, and online services, featuring a total of 8,240 entries with a distribution of 991 negative, 6,613 neutral, and 636 positive comments. Findings reveal that the BERT model outshines the Support Vector Machine in accuracy, achieving a 90% success rate versus the latter's 88%.

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